Encoding of Cyclic Codes

Encoding operation using generator polynomial:

- Encoding of an (n, k) cyclic code in systematic form consists of three steps:
  1. Multiply the message polynomial u(X) by X^{n-k}
  2. Divide $X^{n-k}u(X)$ by g(X) to obtain the remainder b(X)
  3. Form the code word $b(X) + X^{n-k}u(X)$
- All these three steps can be accomplished with a division circuit which is a linear (n-k)-stage shift register with feedback connections based on the generator polynomial
- $g(X) = 1 + g_1X + g_2X^2 + \ldots + g_{n-k-1}X^{n-k-1} + X^{n-k}$

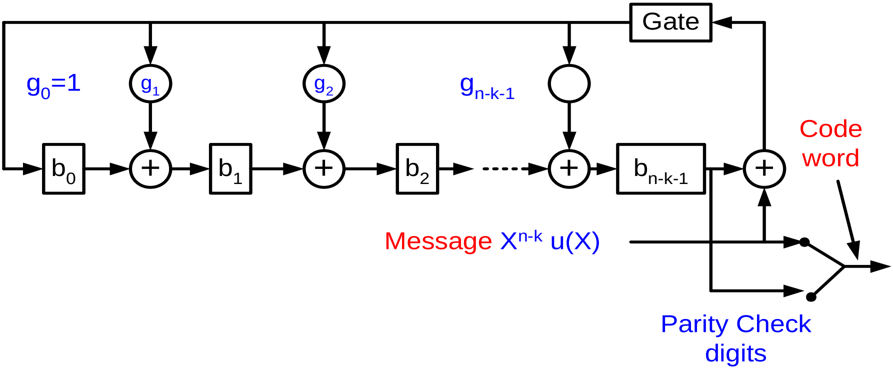

Figure: Encoding circuit for an (n,k) cyclic code

$$g(X) = 1 + g_1X + g_2X^2 + \ldots + g_{n-k-1}X^{n-k-1} + X^{n-k}$$

Encoding operation is as follows:

Step 1

- With the gate turned on, the k information digits $u_0, u_1, \ldots, u_{k-1}$ are shifted into the circuit and simultaneously into the communication channel
- Shifting the message u(X) into the circuit from the front end is equivalent to premultiplying u(X) by X^{n-k}
- As soon as the complete message has entered the circuit, the n - k digits in the register form the remainder and thus they are the parity-check digits.

  Step 2
- Break the feedback connection by turning off the gate.

  Step 3
- Shift the parity-check digits out and send them into the channel
- These n-k parity-check digits $b_0, b_1, \ldots, b_{n-k-1}$, together with the k information digits, form a complete code vector

Example 5.5

- Consider the (7, 4) cyclic code generated by $g(X) = 1 + X + X^3$
- The encoding circuit based on g(X) is shown in Fig. 5.2
- Suppose that the message u = (1 0 1 1) is to be encoded
- As the message digits are shifted into the register, the contents in the register are as follows:

Input	Register contents
	0 0 0 (initial state)
1	1 1 0 (first shift)
1	1 0 1 (second shift)
0	1 0 0 (third shift)
1	1 0 0 (fourth shift)

After four shift, the contents of the register are (1 0 0). The complete codeword is (1 0 0 1 0 1 1)

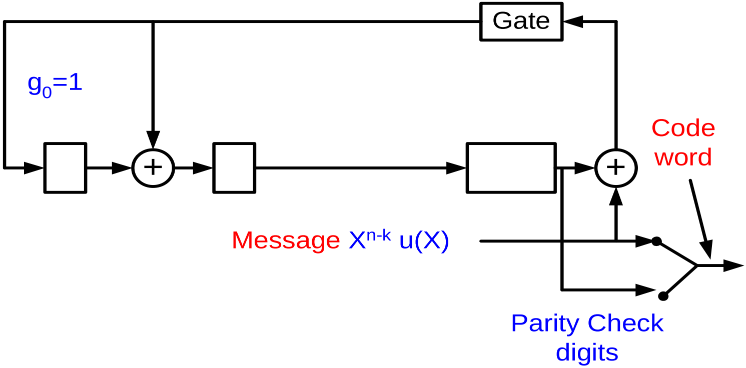

Figure: Encoding circuit for an (n,k) cyclic code

$$g(X) = 1 + g_1X + g_2X^2 + \ldots + g_{n-k-1}X^{n-k-1} + X^{n-k}$$

$$g(X) = 1 + X + X^3$$

Encoding operation using parity polynomial:

- Encoding of a cyclic code can also be accomplished by using its parity polynomial

$$h(X) = h_0 + h_1 X + \ldots + h_k X^k$$

- Let $v = (v_0, v_1, \ldots, v_{n-1})$ be a code vector
- Since $h_k = 1$, the equalities of (5.12) can be put into the following form

$$v_{n-k-j} = \sum_{i=0}^{k-1} h_i v_{n-i-j} \quad for \qquad 1 \leq j \leq n-k \qquad (5.18)$$

- which is known as a difference equation
- Given the k information digits, (5.18) is a rule to determine the n-k parity-check digits, $v_0, v_1, \ldots, v_{n-1}$
- An encoding circuit based on (5.18) is shown in Fig. 5.3

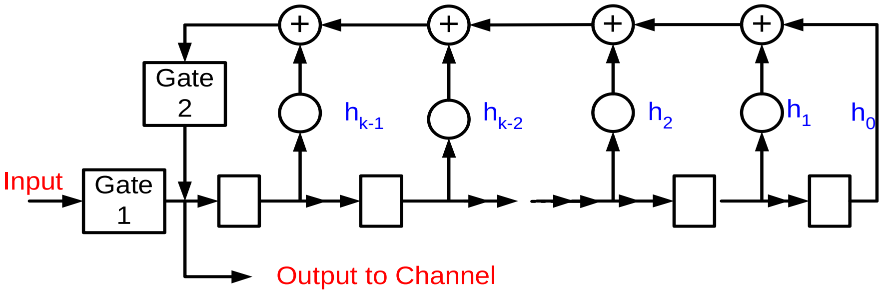

Figure: Encoding circuit for an (n,k) cyclic code

The feedback connections are based on the coefficients of the parity polynomial h(X)

$$h(X) = h_0 + h_1 X + \ldots + h_k X^k$$

The encoding operation can be described in the following steps:

Step 1

- Initially gate 1 is turned on and gate 2 is turned off
- The k information digits $u(X) = u_0 + u_1X + \ldots + u_{k-1}X^{k-1}$ are shifted into the register and the communication channel simultaneously.

  Step 2
- As soon as the k information digits have entered the shift register, gate 1 is turned off and gate 2 is turned on
- The first parity-check digit

$$\begin{aligned} v_{n-k-1} &= h_0 v_{n-1} + h_1 v_{n-2} + \ldots + h_{k-1} v_{n-k} \\ &= u_{k-1} + h_1 u_{k-2} + \ldots + h_{k-1} u_0 \end{aligned}$$

- is formed and appears at point P

Step 3

- The first parity-check digits is shifted
- The second parity-check digits

$$
\begin{aligned}
v_{n-k-2} &= h_0 v_{n-2} + h_1 v_{n-3} + \ldots + h_{k-1} v_{n-k-1} \\
&= u_{k-2} + h_1 u_{k-3} + \ldots + h_{k-2} u_0 + h_{k-1} v_{n-k-1}
\end{aligned}
$$

- is formed and appears at point P.
  Step 4
- Step 3 is repeated until n-k parity-check digits have been formed and shifted into the channel
- Then gate 1 is turned on and gate 2 is turned off
- The next message is now ready to be shifted into the register

Example 5.6

- The parity polynomial of the (7, 4) cyclic code generated by $g(X) = 1 + X + X^3$ is

$$h(X) = (X^7 + 1)/(1 + X + X^3) = 1 + X + X^2 + X^4$$

- The encoding circuit based on h(X) is shown in Fig. 5.4
- The difference equation that determines the parity-check digits is

$$\begin{aligned} v_{3-j} &= 1v_{7-j} + 1v_{6-j} + 1v_{5-j} + 0v_{4-j} \\ &= v_{7-j} + v_{6-j} + v_{5-j} \quad for \quad 1 \leq j \leq 3 \end{aligned}$$

- Suppose that the message to be encoded is (1 0 1 1), then $v_3 = 1, v_4 = 0, v_5 = 1, v_6 = 1$

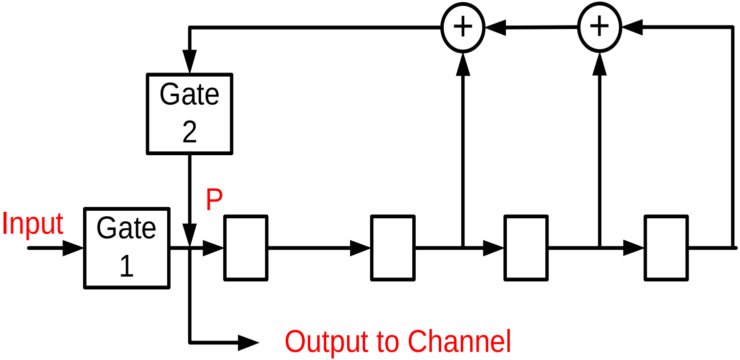

Figure: Encoding circuit for an (n,k) cyclic code

$$h(X) = (X^7 + 1)/(1 + X + X^3) = 1 + X + X^2 + X^4$$

Example 5.6 (cont.)

- The first parity-check digit is $v_2 = v_6 + v_5 + v_4 = 1 + 1 + 0 = 0$
- The second parity-check digit is $v_1 = v_5 + v_4 + v_3 = 1 + 0 + 1 = 0$
- The third parity-check digit is $v_0 = v_4 + v_3 + v_2 = 0 + 1 + 0 = 1$
- The code vector that corresponds to the message (1 0 1 1) is:
- (1 0 0 1 0 1 1)

Input	Register contents
	0 0 0 0 (initial state)
1	1 0 0 0 (first shift)
1	1 1 0 0 (second shift)
0	0 1 1 0 (third shift)
1	1 0 1 1 (fourth shift)
	0 1 0 1 (fifth shift)
	0 0 1 0 (sixth shift)
	1 0 0 0 (seventh shift)

Syndrome Computation and Error Detection

- Let $r = (r_0, r_1, \ldots, r_{n-1})$ be the received vector
- Since the channel noise, the received vector may not be the same as the transmitted code vector
- In the decoding of a linear code, the first step is to compute the syndrome $s = r.H^T$, where H is the parity check matrix
- If the syndrome is zero, r is a code vector and decoder accepts r as the transmitted code vector
- If the syndrome $\neq 0$ r is not a code vector and the presence of errors has been detected

- The received vector r is treated as a polynomial of degree n-1, or less,

$$r(X) = r_0 + r_1X + \ldots + r_{n-1}X^{n-1}$$

- Dividing r(X) by the generator polynomial g(X), we obtain

$$r(X) = a(X)g(X) + s(X) \tag{5.19}$$

- The remainder s(X) is a polynomial of degree n-k-1 or less
- The n - k coefficients of s(X) form the syndrome s
- s(X) is identical to zero if and only if the received polynomial r(X) is a code polynomial.
- The syndrome computation can be accomplished with a division circuit as shown in Fig. 5.5

- The received polynomial r(X) is shifted into the register with all stages initially set to zero.
- As soon as the entire r(X) has been shifted into the register, the contents in the register form the syndrome s(X)

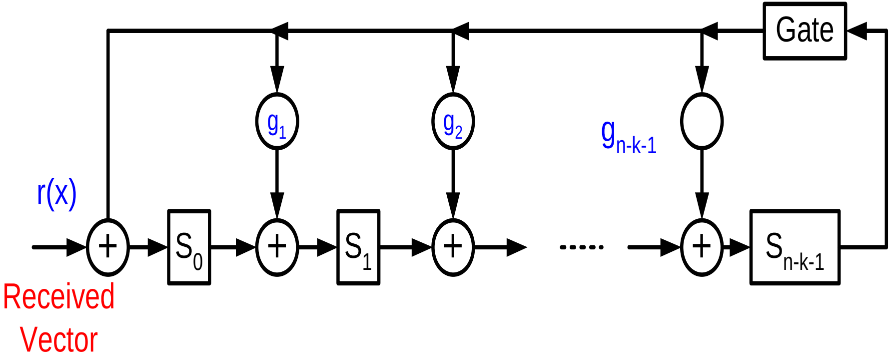

Figure: 5.5: An (n-k) stage syndrome circuit with input from the left end.

Example 5.7

- A syndrome circuit for the (7, 4) cyclic code generated by $g(X) = 1 + X + X^3$ is shown in Fig. 5.6
- Suppose that the received vector is r = (0 0 1 0 1 1 0)
- The syndrome of r is, s = (1 0 1)
- As the received vector is shifted into the circuit, the contents in the register are given in Table 5.3
- At the end of the seventh shift, the register contains the syndrome s = (1 0 1)
- If the register is shifted once more with the input gate disabled, the new contents will be $s^{(1)}(X) = (1\ 0\ 0)$, which is the syndrome of $r^{(1)}(X) = (0\ 0\ 0\ 1\ 0\ 1\ 1)$, a cyclic shift of r

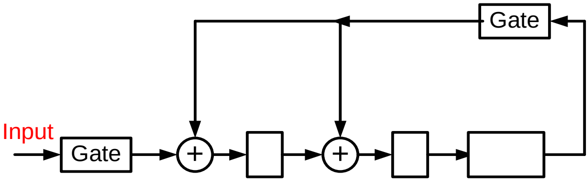

Figure: 5.6: Syndrome circuit for (7, 4) cyclic code generated by $g(x) = 1 + x + x^3$

Table: 5.3: Contents of the syndrome register with r=(0010110)and r=(1011011)

Shift	Input	Register contents		
		0 0 0 (initial state)		0 0 0
1	0	0 0 0 (first shift)	1	1 0 0
2	1	1 0 0 (second shift)	1	1 1 0
3	1	1 1 0 (third shift)	0	0 1 1
4	0	0 1 1 (fourth shift)	1	0 1 1
5	1	0 1 1 (fifth shift)	1	0 1 1
6	0	1 1 1 (sixth shift)	0	1 1 1
7	0	1 0 1 (syndrome s)	1	0 0 1
8	-	1 0 0 (syndrome $s^{(1)}$)		
9	-	0 1 0 (syndrome $s^{(2)}$)		

Syndrome computation by giving the input from the right end.

- Shift the received vector r(X) into the syndrome register from the right end, as shown in Fig. 5.7
- The content of the register do not form the syndrome of r(X) but it form the syndrome $s^{(n-k)}(X)$ of $r^{(n-k)}(X)$, which is the $(n-k)^{th}$ cyclic shift of r(X).
- Shifting r(X) from the right end is equivalent to pre-multiplying r(X) by X^{n-k}
- When the entire r(X) has entered the register, the register contains the remainder $\rho(X)$ resulting from dividing $X^{n-k}r(X)$ by the generator polynomial g(X)
- Thus, we have

$$X^{n-k}r(X) = a(X)g(X) + \rho(X) \tag{5.24}$$

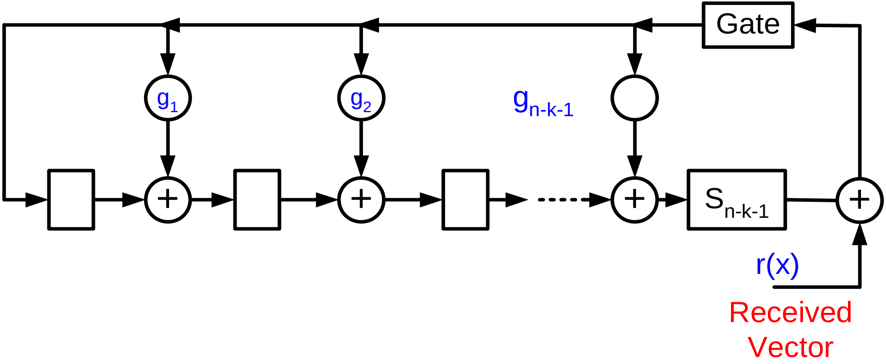

Figure: 5.7: An (n-k) syndrome circuit with input from the right end

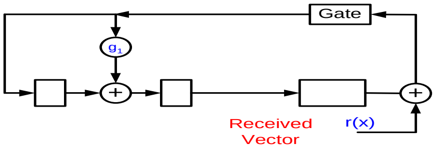

Figure: 5.71: An (n-k) syndrome circuit with input from the right end

Left shift r(x) by (n-k) times

0010110 => 0001011 => 1000101 => 1100010

Table: 5.3: Contents of the syndrome register with r=(1100010)

Shift	Input	Register contents
		0 0 0 (initial state)
1	0	0 0 0 (first shift)
2	1	1 1 0 (second shift)
3	0	0 1 1 (third shift)
4	0	1 1 1 (fourth shift)
5	0	1 0 1 (fifth shift)
6	1	0 1 0 (sixth shift)
7	1	1 0 1 (syndrome s)

- It follows from (5.1) that r(X) and $r^{(n-k)}(X)$ satisfy the following relation:

$$X^{n-k}r(X) = b(X)(X^n + 1) + r^{(n-k)}(X) \tag{5.25}$$

- Combining (5.24) & (5.25) and using the fact that $X^n + 1 = g(X)h(X)$
- We have $r(n-k)(X) = [b(X)h(X) + a(X)]g(X) + \rho(X)$
- Therefore,$\rho(X)$ is indeed the syndrome of $r^{(n-k)}(X)$
- Let v(X) be the transmitted code word and let $e(X) = e_0 + e_1X \ldots + e_{n-1}X^{n-1}$ be the error pattern.

- The received polynomial is $r(X) = v(X) + e(X)$
- Since v(X) is a multiple of the generator polynomial g(X), we have: $e(X) = [a(X) + b(X)]g(X) + s(X)$
- where $b(X)g(X) = v(X)$.
- This shows that the syndrome is actually equal to the remainder resulting from dividing the error pattern by the generator polynomial.

Decoding of Cyclic Codes

General cyclic(Meggitt) decoder.

- Decoding of cyclic code consists of following three steps:
  1. Syndrome computation
  2. Association of the syndrome to an error pattern
  3. Error correction
- The syndrome computation for cyclic codes can be accomplished with a division circuit.
- A straightforward approach to the design of a decoding circuit is via a combinational logic circuit that implements the table-lookup procedure.
- The limit to this approach is that the complexity of the decoding circuit tends to grow exponentially with the code length and the number of errors that we intend to correct.
- The cyclic structure of a cyclic code allows us to decode a received vector $r(X) = r_0 + r_1X + r_2X^2 + \ldots + r_{n-1}X^{n-1}$ in a serial manner.
- The received digits are decoded one at a time and each digit is decoded with the same circuitry.

- As soon as the syndrome has been computed, the decoding circuit checks whether the syndrome s(X) corresponds to a correctable error pattern $e(X) = e_0 + e_1X + \ldots + e_{n-1}X^{n-1}$ with an error at the highest-order position $X^{n-1}(i.e., e_{n-1} = 1)$.
- If s(X) does not correspond to an error pattern with $e_{n-1} = 1$, the received polynomial and the syndrome register are cyclically shifted once simultaneously.
- By doing so, we obtain $r^{(1)}(X) = r_{n-1} + r_0X + r_1X^2 + \ldots + r_{n-2}X^{n-1}$ and the new contents in the syndrome register form the syndrome $s^{(1)}(X)$ of $r^{(1)}(X)$.
- The same decoding circuit will check whether $s^{(1)}(X)$ corresponds to an error pattern with an error at location X^{n-1}.
- If the syndrome s(X) does correspond to an error pattern with $e_{n-1} = 1$, the first received digit r_{n-1} is an erroneous digit and it must be corrected.
- This correction is carried out by $r_{n-1} \oplus e_{n-1}$.

- This correction results in a modified received polynomial $r_1(X) = r_0 + r_1X + r_2X^2 + \ldots + ({}_{rn-1} \oplus e_{n-1})X^{n-1}$.
- The effect of the error digit $e^n - 1$ on the syndrome is then removed from the syndrome s(X).
- This can be achieved by adding the syndrome of $e^{'}(X) = X^{n-1}$ to s(X).
- This sum is the syndrome of the modified received polynomial $r_1(X)$.
- Cyclically shift $r_1(X)$ and the syndrome register once simultaneously.
- This shift results in a received $r_1^{(1)}(X) = (r_{n-1} \oplus e_{n-1}) + r_0X + \ldots + r_{n-2}X^{n-1}$.

- The syndrome $s_{(1)}^{1}(X)$ of $r_{(1)}^{1}(X)$ is the remainder resulting from dividing $X[s(X) + X^{n-1}]$ by the generator polynomial g(X).
- Since the remainders resulting from dividing $X_s(X)$ and X^n by g(X) are $s^{(1)}(X)$ and 1, respectively, we have $s_1^{(1)}(X) = s^{(1)}(X) + 1$.
- Therefore, if 1 is added to the left end of the syndrome register while it is shifted, we obtain $s_{(1)}^{1}(X)$.
- A general decoder for an (n, k) cyclic code is shown in Fig. 5.8
- It consists of three major parts:
    1. A syndrome register
    2. An error-pattern detector
    3. A buffer register to hold the received vector
- To remove the effect of an error digit on the syndrome, we simply feed the error digit into the shift register from the left end through an EXCLUSIVE-OR gate

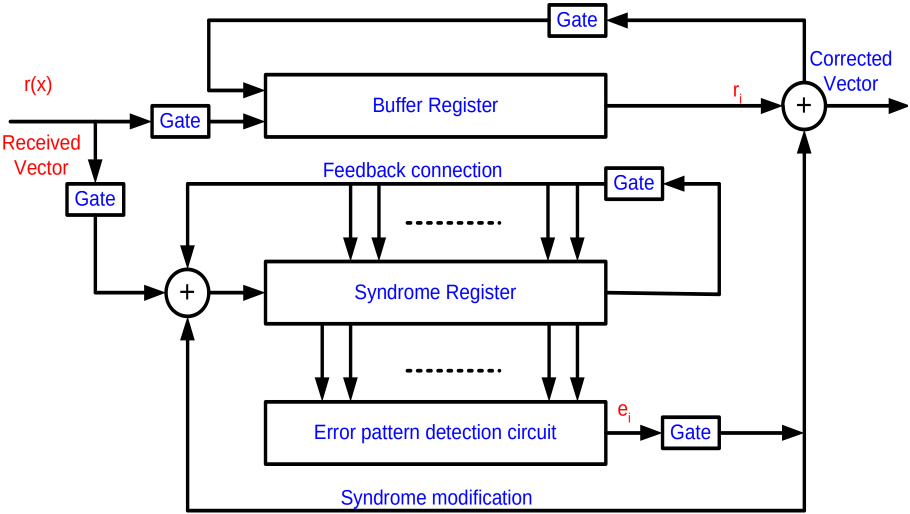

Figure: General cyclic(Meggitt) decoder with received polynomial r(X) is shifted into the syndrome register from the left end.

- The decoding operation is described as follows:
  Step 1
- The syndrome is formed by shifting the received vector into the syndrome register & also the received vector is stored into the buffer register.
  Step 2
- The syndrome is read into the detector and is tested for the corresponding error pattern.
- The detector is a combinational logic circuit which is designed in such a way that its output is 1 iff the syndrome in the syndrome register corresponds to a correctable error pattern with an error at the highest-order position X^{n-1}
- If a “1” appears at the output of the detector, the received symbol in the rightmost stage of the buffer register is assumed to be erroneous, if a “0” appears, the received symbol at the rightmost stage of the buffer register is assumed to be correct and no correction necessary.
- The output of the detector is the estimated error value for the symbol to come out of the buffer

Step 3

- The first received symbol is read out of the buffer
- If the first received symbol is detected to be an erroneous symbol, it is corrected by the output of the detector
- The output of the detector is fed back to the syndrome register to modify the syndrome
- This results in a new syndrome, which corresponds to the altered received vector shifted one place to the right.

Step 4

- The new syndrome formed in step 3 is used to detect whether or not the second received symbol is an erroneous symbol.
- The decoder repeats step 2 and 3

Step 5

- The decoder decodes the received vector symbol by symbol in the manner outlined above until the entire received vector is read out of the buffer register
- The decoder above is known as Meggitt decoder

Example 5.9

- Consider the decoding of the (7, 4) cyclic code generated by $g(X) = 1 + X + X^3$
- This code has minimum distance 3 and is capable of correcting any single error over a block of seven digits
- There are seven single-error patterns
- These seven error patterns and the all-zero vector form all the coset leader of the decoding table
- They form all the correctable error patterns
- Suppose that the received polynomial $r(X) = r_0 + r_1X + r_2X^2 + \ldots + r^6X^6$
- is shifted into the syndrome register from the left end
- The seven single-error patterns and their corresponding syndromes are listed in Table 5.4

Table: 5.4: Error patterns & their syndromes with received polynomial r(X) shifted into the circuit from the left end.

Error Pattern	Syndrome	Syndrome vector
$e_6(x) = X^6$	$s(x) = 1 + X^2$	(101)
$e_5(x) = X^5$	$s(x) = 1 + X + X^2$	(111)
$e_4(x) = X^4$	$s(x) = X + X^2$	(011)
$e_3(x) = X^3$	$s(x) = 1 + X$	(110)
$e_2(x) = X^2$	$s(x) = X^2$	(001)
$e_1(x) = X^1$	$s(x) = X$	(010)
$e_0(x) = X^0$	$s(x) = 1$	(100)

Table: 5.5: Error patterns & their syndromes with received polynomial r(X) shifted into the circuit from the right end.

Error Pattern	Syndrome	Syndrome vector
$e_6(x) = X^6$	$s^{(3)}(x) = X^2$	(001)
$e_5(x) = X^5$	$s^{(3)}(x) = X$	(010)
$e_4(x) = X^4$	$s^{(3)}(x) = 1$	(100)
$e_3(x) = X^3$	$s^{(3)}(x) = 1 + X^2$	(101)
$e_2(x) = X^2$	$s^{(3)}(x) = 1 + X + X^2$	(111)
$e_1(x) = X^1$	$s^{(3)}(x) = X + X^2$	(011)
$e_0(x) = X^0$	$s^{(3)}(x) = 1 + X$	(110)

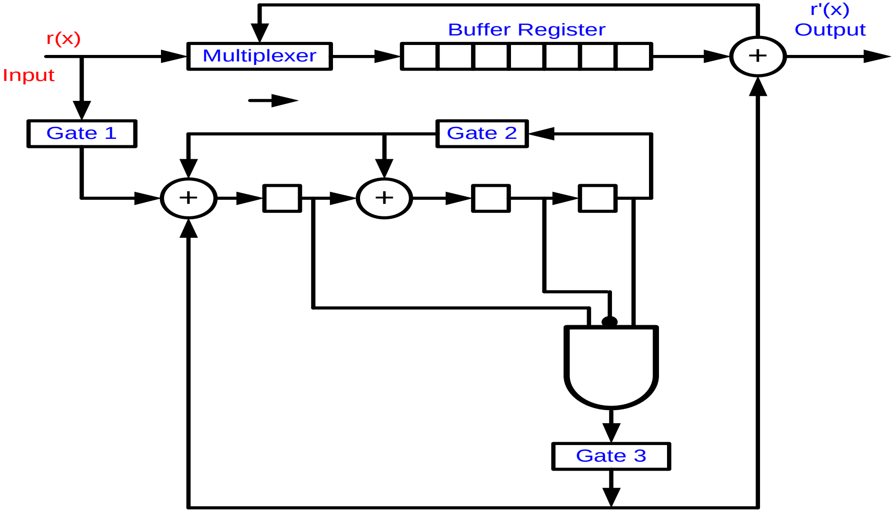

Figure: Decoding circuit for the (7,4) cyclic code generated by $g(x) = 1 + x + x^2$.

Table: 5.41: Contents of the syndrome register with r=(1001010) error at location X^6

Shift	Input	Register contents			Shift
	Input	$s_0 = i_0 \oplus s_2^{(-1)}$	$s_1 = s_0 \oplus s_2^{(-1)}$	$s_2 = s_1^{(-1)}$	
		0	0	0	(initial state)
1	0	0	0	0	(first shift)
2	1	1	0	0	(second shift)
3	0	0	1	0	(third shift)
4	1	1	0	1	(fourth shift)
5	0	1	0	0	(fifth shift)
6	0	0	1	0	(sixth shift)
7	1	1	0	1	(syndrome s)

Table: 5.42: Contents of the syndrome register with r=(1001001) error at location X^5

Shift	Input	Register contents			Shift
	Input	$s_0 = i_0 \oplus s_2^{(-1)}$	$s_1 = s_0 \oplus s_2^{(-1)}$	$s_2 = s_1^{(-1)}$	
		0	0	0	(initial state)
1	1	1	0	0	(first shift)
2	0	0	1	0	(second shift)
3	0	0	0	1	(third shift)
4	1	0	1	0	(fourth shift)
5	0	0	0	1	(fifth shift)
6	0	1	1	0	(sixth shift)
7	1	1	1	1	(syndrome s)

Table: 5.31: Contents of the syndrome register with r=(1011011)error at location X^2

Shift	Input	Register contents			Shift
	Input	$s_0 = i_0 \oplus s_2^{(-1)}$	$s_1 = s_0 \oplus s_2^{(-1)}$	$s_2 = s_1^{(-1)}$	
		0	0	0	(initial state)
1	1	1	0	0	(first shift)
2	1	1	1	0	(second shift)
3	0	0	1	1	(third shift)
4	1	0	1	1	(fourth shift)
5	1	0	1	1	(fifth shift)
6	0	1	1	1	(sixth shift)
7	1	0	0	1	(syndrome s)

Table: 5.32: Contents of the syndrome register with r=(1011011)error at location X^6 after 4 shifts

Shift	Register contents			Shift
	$s_0 = i_0 \oplus s_2^{(-1)}$	$s_1 = s_0 \oplus s_2^{(-1)}$	$s_2 = s_1^{(-1)}$	
	0	0	1	(initial state)
1	1	1	0	(first shift)
2	0	1	1	(second shift)
3	1	1	1	(third shift)
4	1	0	1	(fourth shift)
5	0	0	0	(fifth shift)
6	0	0	0	(sixth shift)
7	0	0	0	(syndrome s)